Unified Memory Showdown · 128GB+ Local-AI Boxes

Price / availability / pros & cons for local LLM work — what to buy in the “RAMpocalypse”

Compiled Oct 14, 2026

System	Chip	Memory	Bandwidth	Price (USD)	Availab
NVIDIA DGX Spark	GB10 Grace Blackwell (Arm)	128GB unified (~96GB to GPU)	NVLink C2C — highest effective in class	**$4,699 → reported hike to $6,950 (Oct 2); 64GB SKU at $4,999**	Availabl (Micro Center, Best Bu pricing unstabl
AMD Ryzen AI Halo 192GB	Ryzen AI Max+ PRO 495 (x86)	192GB LPDDR5X-8533 (160GB GPU-assignable)	273 GB/s	**$7,299.99**	Now at Micro Center; Linux o (Windov “Project Zenith” TBD)
AMD Ryzen AI Halo 128GB	Ryzen AI Max+ 395	128GB	256 GB/s	**$3,999.99**	Availabl
Apple Mac Studio M5 Ultra	M5 Ultra, 36c CPU / 80c GPU	96GB base; 256GB now; 512GB “late Oct”	1.2 TB/s	**$5,499 base; 256GB +$4,000 (~$9,499); 512GB TBD (full spec >$20K)**	Shippin since Se 22; 512 config withhel can’t ev pre-ord
Framework Desktop 128GB	Ryzen AI Max+ 395	128GB	256 GB/s	**$3,449 (192GB PRO 495 ver: $6,799)**	Spotty - out of s Oct 1; repairab open design
Minisforum MS-S1 MAX	Ryzen AI Max+ 395	128GB LPDDR5x	256 GB/s	**$3,879**	In stock (minisfo store); 10GbE

Pros & Cons

DGX Spark — speed champ under 128GB

+ ~30% faster tokens than AMD Halo on sub-64GB models (NVLink C2C bandwidth)
+ CUDA = deepest ecosystem; preloaded NVIDIA AI stack; up to 1 PFLOP FP4
− 128GB hard ceiling, no 192GB path; price swing $4,699→$6,950
− Arm + Linux workflow; nothing suggests prices dropping near-term

Ryzen AI Halo 192GB — capacity king

+ Only compact box that loads 300B-class models at INT4 (~150GB weights)
+ x86; ROCm 10.0 validates vLLM, llama.cpp, Ollama, LM Studio
− ~30% slower than Spark below ~64GB; Linux-only at launch
− KV-cache headroom tight at long context; ROCm still trails CUDA in long tail

Mac Studio M5 Ultra — raw bandwidth

+ 1.2 TB/s — fastest raw token throughput in this table
+ Fully usable daily-driver Mac; 512GB option lands late Oct
− Apple RAM tax: 96→256GB = $4,000; 512GB price unannounced
− MLX/Metal ecosystem, not CUDA; supply-shortage games on big configs

Framework / Minisforum / Beelink / GMKtec — value picks

+ Same Strix Halo silicon from $3,449–3,879 — cheapest 128GB unified
+ Framework: repairable, open; Minisforum: dual 10GbE
− 256 GB/s = bandwidth-bound, inherently slower generation than Spark
− Availability volatile; same ROCm caveats as Halo

Bottom line

- **≤70B models, best all-around:** DGX Spark — but only if you land it at $4,699, not $6,950
- **Cheapest 128GB ticket:** Framework Desktop $3,449 (when stocked); Minisforum MS-S1 MAX $3,879 backup
- **200B+/300B-class:** 192GB Halo at $7,300 — slower per token, but nothing else fits the weights
- **Fastest inference + real workstation:** M5 Ultra Mac Studio, if you swallow Apple’s RAM pricing

Context

These win over desktop DDR5 rigs because the **entire pool is GPU-addressable at high bandwidth** — capacity alone doesn’t replicate that. 2026 DRAM/HBM shortage is repricing the whole category upward; buy sooner rather than later if you need one.

Sources: NVIDIA newsroom & marketplace; Apple Newsroom (Aug 25 / Sep 22, 2026) & AppleInsider; TechTimes (Oct 14, 2026) citing HotHardware benchmarks; shattered.io / Hardware Busters / Tom’s Hardware (Oct 2, 2026); Phoronix & HWBusters (Sep 30, 2026); frame.work, Micro Center, Minisforum store listings. Prices as listed Oct 14, 2026 — verify before purchase; DGX Spark 128GB pricing conflicts between $4,699 (retail listings) and $6,950 (reported repricing).